

UPI Transaction Risk Profiler

Fraud Detection & Behavioral Risk Analysis

Python · MySQL · Power BI · ReportLab

Business Problem

UPI processed 17,220 crore transactions worth RS. 246 lakh crore in FY2024-25, making India the world's largest real-time payments market by volume. With fraud cases rising, banks need automated behavioral risk detection systems that go beyond static rules. This project builds a Fraud Velocity Index (FVI) to detect anomalous UPI payment behavior and flag high-risk transactions before loss occurs.

Dataset Overview

Property	Detail
Source	PaySim Synthetic Financial Dataset (Kaggle)
Total Size	6.3 million transactions, 11 features
Fraud Rate	0.13% — 8,213 fraudulent transactions
Fraud Types	TRANSFER and CASH_OUT only
Sample Used	50,000 rows loaded into MySQL & Power BI

Tools & Technologies

Tool	Purpose
Python	EDA, feature engineering, visualization
MySQL	Database design, SQL analysis queries
Power BI	Interactive dashboard, What-If simulator
ReportLab	PDF summary report generation
GitHub	Version control, project documentation

Fraud Velocity Index (FVI)

FVI is a custom behavioral risk score created for this project, combining three transaction signals into a single risk number (0–6 scale).

Signal	Logic	Weight
Balance Mismatch	Account balances do not reconcile	2 points
Zero Drain	Account wiped to zero after transaction	3 points
Amount Ratio	Transaction large relative to account size	0–1 points

Key Findings

Finding 1 — Fraud concentrated in 2 transaction types

100% of fraud occurs in TRANSFER (0.84% rate) and CASH_OUT (0.20% rate). PAYMENT, DEBIT, and CASH_IN show zero fraud. Risk monitoring can focus exclusively on these two types, reducing screening cost by 60%+.

Finding 2 — Fraud peaks at specific hours

Fraud spikes around step 300 with 3x higher volume than average hours. Automated monitoring and alerts should be strongest during this peak window when human oversight is lowest.

Finding 3 — FVI successfully separates fraud from normal

Critical Risk segment ($FVI \geq 4.0$) contains 12,012 transactions and captures 95.7% of all actual fraud. Low Risk segment ($FVI < 1.5$) contains 7,305 transactions with zero fraud — these can be cleared instantly with no review.

Finding 4 — High value, single-use fraud accounts

Top fraud accounts each transferred exactly RS. 1 crore in a single transaction. Every fraud account was used exactly once, making traditional blacklisting ineffective. Behavioral scoring is the correct detection approach.

FVI Risk Segmentation Results

Risk Segment	Transactions	Actual Fraud	Fraud Rate
Critical Risk ($FVI \geq 4$)	12,012	67	0.56%
High Risk ($FVI 3-4$)	20,134	1	0.00%
Medium Risk ($FVI 1.5-3$)	10,549	2	0.02%
Low Risk ($FVI < 1.5$)	7,305	0	0.00%

Business Recommendations

Recommendation 1 — Focus screening on TRANSFER and CASH_OUT

Flag all TRANSFER and CASH_OUT transactions above RS. 50,000 for enhanced automated screening. These are the only two types with confirmed fraud activity, allowing risk teams to eliminate unnecessary checks on 3 other transaction types entirely.

Recommendation 2 — Set FVI operational threshold at 4.0

An FVI threshold of 4.0 captures 95.7% of fraud while limiting review workload to 24% of total transactions. This is the optimal balance between detection sensitivity and operational cost. The Power BI What-If simulator allows risk managers to adjust this threshold in real time as fraud patterns evolve.

Recommendation 3 — Heighten monitoring during peak fraud hours

Implement enhanced automated alerts during step 280–320 (equivalent to peak fraud window). Fraud volume during this period is 3x above daily average. Deploying additional rule-based checks during this window can intercept fraud before settlement occurs.

Recommendation 4 — Replace blacklisting with behavioral scoring

Since every fraud account in this dataset was used exactly once, account-level blacklisting has zero effectiveness against new fraud attempts. FVI-based transaction-level behavioral scoring is the correct long-term approach, consistent with RBI's 2024 revised guidelines on early warning systems.